

Math 421

Wednesday, November 26

Announcements

- This week's drop-in hours:
 - **Today** 12-1 pm VV 311
- Check for any final exam conflicts & email me
- Have a great break!

Chain rule

Theorem 28.4. If f is differentiable at a and g is differentiable at $f(a)$, then $g \circ f$ is differentiable at a and

$$(g \circ f)'(a) = g'(f(a)) \cdot f'(a)$$

“Proof” of Chain Rule

Q: What is wrong with this proof?

Chapter 5: Differentiation

Section 29: The Mean Value Theorem

Motivating Questions

Question: How do we rigorously show the following?

- ❖ If $f'(x) = 0$ for all x , then f is constant.
- ❖ If $f'(x) > 0$ for all x , then f is strictly increasing.
- ❖ If $f'(x) < 0$ for all x , then f is strictly decreasing.

Answer: The Mean Value Theorem!

Extreme Values

Theorem 29.1. If f is defined on an open interval containing x_0 , if f assumes its maximum or minimum at x_0 , and if f is differentiable at x_0 , then $f'(x_0) = 0$.

Discussion Questions – True/False

- 1 If $f'(x) = 0$, then x_0 is a maximum or a minimum.
- 2 If x_0 is a maximum or a minimum, then $f'(x) = 0$.

Rolle's Theorem

Theorem 29.2. Let f be a continuous function on $[a, b]$ that is differentiable on (a, b) and satisfies $f(a) = f(b)$. There exists (at least one) $x \in (a, b)$ such that $f'(x) = 0$.

Discussion.

Rolle's Theorem

Theorem 29.2. Let f be a continuous function on $[a, b]$ that is differentiable on (a, b) and satisfies $f(a) = f(b)$. There exists (at least one) $x_0 \in (a, b)$ such that $f'(x_0) = 0$.

Proof.

Mean Value Theorem

Theorem 29.3. If f is continuous on $[a, b]$ and differentiable on (a, b) , then there is a number $x_0 \in (a, b)$ such that

$$f'(x_0) = \frac{f(b) - f(a)}{b - a}$$

Proof of MVT

Proof.

Constant Functions

Corollary 29.4. If f is defined on (a, b) and $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on (a, b) .

True/False: If $f'(x) = 0$ for all $x \in \text{dom}(f)$ then f is constant.

Proof of Corollary

Corollary 29.4. If f is defined on (a, b) and $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on (a, b) .

Proof.

Activity – Prove the Corollary

Corollary 29.5. If f and g are defined on (a, b) and $f'(x) = g'(x)$ for all $x \in (a, b)$, then there is some number c such that $f(x) = g(x) + c$ for all $x \in (a, b)$.

Proof.

Increasing & Decreasing

Def. Let f be a real-valued function defined on an interval I . If $x_1, x_2 \in I$ and $x_1 < x_2$ we say

1. f is **increasing** on I if $f(x_1) \leq f(x_2)$ and
2. f is **strictly increasing** on I if $f(x_1) < f(x_2)$

Examples:

Another Corollary of the MVT

Corollary 29.7. Let f be a differentiable function on an interval (a, b) . Then

1. f is increasing if $f'(x) \geq 0$ for all $x \in (a, b)$.
2. f is strictly increasing if $f'(x) > 0$ for all $x \in (a, b)$.

Q: How does this follow from the MVT?

Discussion Question

Is the derivative of a differentiable function always continuous?

Example

Let

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

You prove

$$f'(x) = \begin{cases} -\cos \frac{1}{x} + 2x \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

IVT for Derivatives

Thm. Let f be a differentiable function on (a, b) . Then $f'(x)$ has the intermediate value property on (a, b) . That is, if $a < x_1 < x_2 < b$, and if c lies between $f'(x_1)$ and $f'(x_2)$, there exists (at least one) x in (x_1, x_2) such that $f'(x) = c$.

Activity – Prove the Theorem

Step 1: Define $g(x) = f(x) - cx$. Why does this make sense as a choice?

Step 2: Assume $f'(x_1) < c < f'(x_2)$. Show that $g(x)$ assumes its minimum at some point $x_0 \in (x_1, x_2)$. Why can't the minimum occur at the endpoints?

Step 3: Get that $g'(x_0) = 0$ and show that this gives the desired conclusion.